

Laboratory Work No. 4

Aggregation and Grouping in SQL

Objective:

Practice SQL aggregate functions and grouping techniques for summarizing and analyzing data stored in a relational database.

Introduction:

This laboratory focuses on the use of SQL aggregate functions (COUNT, SUM, AVG, MIN, MAX), grouping (GROUP BY), and conditional filters (HAVING) to generate summarized reports from relational data. Through these exercises, you will learn how to produce department-level summaries, capacity analyses, and budget comparisons.

Task:

Using your existing database, write SQL queries for each of the following tasks.

1. Show the total number of courses and the average course credit value across all courses.
2. For the Spring 2024 semester, find how many distinct courses are offered, and the total number of sections scheduled.
3. List each department with the number of instructors and the average salary, sorted by average salary in descending order.
4. List each department with the number of students and their average total credits, sorted by student count in descending order.
5. Show each building with the number of sections offered in Fall 2023 and the average classroom capacity.
6. List the departments where the average student total credits are greater than 25.
7. Show the departments that have more than 5 students and their average total credits.
8. Show the departments where the total instructor salary is greater than the department budget.

Deliverables:

1. A report containing your SQL queries and their outputs.
2. Include clear screenshots showing both the query and its result.
3. Ensure that all results are properly labeled, well formatted, and easy to read.

Good luck!